

Changing a Networked Mind: Paradigm Dynamics as Structure Learning over a Hidden World

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Abstract

A scientist is not handed the variables of the world, nor the dependencies among them: the structure of reality is hidden, and a community must *learn* that structure even as it uses it. We model a paradigm as a directed network of commitments and a scientific community as a population of bounded active-inference agents performing *Bayesian structure learning* on that network—expanding it by proposing new commitments and reducing it by pruning the dependencies the data do not support. One operator organizes the formalism: the propagation operator of the network carries a *conservatism* field (the cost of revising a commitment) and a *conviction* field (the value a bounded agent attaches to holding it), the same operator applied to two different sources. Running the model through the full Kuhnian cycle—normal science, anomaly, crisis, reduction, discovery—in a 1,816-run sweep yields three findings. The lock-in boundary sits at *very* weak social coupling: conviction-driven gating of disconfirming evidence cannot defeat connection, and only isolation protects a superseded paradigm. Discovery of a genuinely unconceived dimension follows a Poisson waiting-time law, but a *staggered* discovery does not survive the communal prior—survival is zero at slow proposal rates and appears only once arrivals cluster the community’s wakings within a few steps of one another, a critical mass in *time*. And the predicted belt-first/core-last staircase is a creature of *observability*, not of a revision-rate gradient: it appears when the core commitments carry no direct observation rows—evidence reaching them only through relational channels—and vanishes under full observability, which localizes what a richer model must add to data-dependent couplings.

1 Introduction: changing a frame, not a credence

To change one’s mind about a fact is to update a belief; to change a paradigm is to change the *frame* in which beliefs are posed. Such a frame is not a credence but a *structure*: a network of conditional dependencies through which information propagates. A community that only revised its credences, leaving the network fixed, could not change frame at all. This is the phenomenon Kuhn named but could not run forward (Kuhn, 1962): nothing in *The Structure of Scientific Revolutions* tells us, mechanistically, why a community holds a structure of commitments long past the first disconfirming evidence, nor what finally makes the holding give way.

When the chemical community released phlogiston it did not revise a single credence. It rewrote a network of commitments in which combustion, calcination, respiration, and the theory of acids were all load-bearing on a hub that had to go: late-phlogistic theorists proposed cascading revisions that proved mutually incompatible (Blumenthal and Ladyman, 2017; Holmes, 2000), defending less a hypothesis about combustion than a constellation of entangled commitments (Boantz and Gal, 2011)—phlogiston was load-bearing for too much.

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The dominant formalism for such communities, network epistemology, cannot represent this case. Its agents hold a scalar credence in a fixed proposition and update on a trust graph (Zollman, 2010; O’Connor and Weatherall, 2019; Zollman, 2013), a programme that has produced striking results on transient diversity, polarization, and the division of cognitive labour (Jonard et al., 2025; Michelini et al., 2023; Kitcher, 1990; Strevens, 2003)—but the phlogiston puzzle is not a failure to converge on a number. The commitments were *entangled*: revising any one forced a re-reading of the others, and the disagreement was about which entanglements to keep. Thagard’s ECHO (Thagard, 1989, 1990) captures the structural intuition as undirected coherence; it lacks the directed dependencies, the carry-over cost of mind-change, and the expand/reduce asymmetry we derive below. Active-inference accounts of epistemic communities supply an endogenous precision-based lock-in mechanism (Albarracín et al., 2022) and warn that naive aggregation degrades collective precision (Friston et al., 2024a; Waade et al., 2025); the variational cost of mind-change has been formalized for a single agent as a motivated trade between revision cost and the utility of a held belief (Hyland and Albarracín, 2025). What none of these does is treat *belief structure itself* as the object of inference: they retain a fixed menu of hypotheses, and a scalar credence over a menu cannot say how revising one commitment forces a re-reading of the rest, nor how a community arrives at a *new* way of carving up the world. When only a number moves on the trust graph, a community can converge on it, herd on it, or split over it—but its members can never come to hold different *kinds* of world, only different values on a shared one. The question, made concrete in §3.5, is therefore what moves: this paper lets a piece of *structure* travel, so that one agent can come to represent a commitment its peers do not.

We make structure the object. Agents carry a directed network of commitments, learn its structure by Bayesian model expansion and reduction, and are *bounded*: they sample a complex world through narrow channels, value some conclusions over others, and judge what they receive against their own model. Section 3 builds the model and states the simulation loop explicitly; Section 4 reports a single-agent dark-energy net and a 1,816-run population sweep through the full Kuhnian cycle on the phlogiston world.

2 Why paradigms are networks under structure learning

The structure of reality is hidden. A scientist confronting a new domain is not given the right variables, and is certainly not given the dependencies among them: she does not know whether two regularities share a hidden common cause, are lawfully coupled, or merely coincide. For an active-inference reader the move is natural: just as agents infer hidden states, they must infer the hidden *dependencies among* those states. The cognitive-science literature on theory acquisition reaches the same conclusion—conceptual development as hierarchical Bayesian program induction, a stochastic search over structures (Ullman and Tenenbaum, 2020; Kemp and Tenenbaum, 2008; Ullman et al., 2012).

To explore is to expand; to do normal science is to reduce. Network-epistemology models keep their communities from collapsing onto the first attractive theory with a hand-tuned exploration term—an ϵ , a temperature, a switching cost (Zollman, 2010; O’Connor and Weatherall, 2019). Once structure is the object of inference, the *content* of exploration is no longer a tuned bonus but the epistemic value of expanding the model to entertain a commitment not yet in it (Friston et al., 2015, 2017); what stays free is only the *rate* at which proposals arrive. The complementary operation is Bayesian model reduction: pruning structure the data do not support, in closed form, at the slowest timescale of the learning hierarchy (Friston et al., 2018; Smith et al., 2022b). Existing applications

of this machinery run on single agents (Smith et al., 2022b; Priorelli and Stoianov, 2023) or define “group” as between-subject neuroimaging (Friston et al., 2016); treating a paradigm as the object a *community* jointly reduces and expands is the open move this paper makes.

Two fields, one operator. The single non-obvious step the rest of the paper turns on: once a paradigm is a network, one operation that network science already owns (§3.1)—propagating influence from a commitment to everything downstream—generates *two* fields on the same graph, depending on what is propagated. Propagate sheer dependence and we get *conservatism*: a commitment is costly to revise because much depends on it. Propagate *value* and we get *conviction*: the worth a bounded agent attaches to holding a belief, carried through the same dependencies. The two are independent—a commitment can be central yet value-neutral, cheap yet cherished (Fig. 1). That attitudes live on a network is the empirical posture of the Causal Attitude Network model (Dalege et al., 2016); we place it on the same *directed* graph that carries the cost of revision.

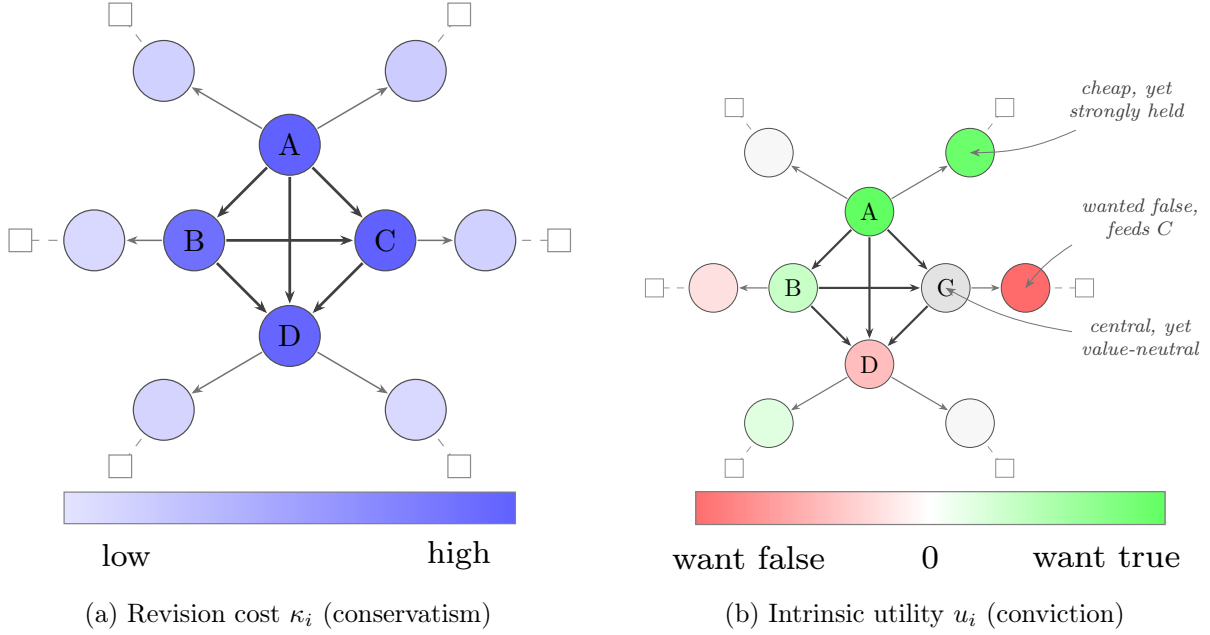


Figure 1: Two fields on one paradigm (schematic). Both panels carry the *same* network—a densely coupled core $\{A, B, C, D\}$ and a sparse belt $\{b_1, \dots, b_6\}$ in contact with observation. a Revision cost is the carry-over of mind-change, $\kappa = \mathbf{T}\mathbf{1}$ (Eq. 1): a node’s cost is the precision-weighted mass of its descendants, a clean radial gradient monotone in centrality. b Intrinsic utility is signed—beliefs one wants true (green), wants false (red), or is neutral toward (white)—and the conviction field $U = \mathbf{T}u$ does not align with κ : C is central yet value-neutral; b_2 is cheap yet strongly held; b_4 , which feeds C , is a conclusion the community wants false. The two fields are $\mathbf{T}\mathbf{1}$ and $\mathbf{T}u$, the same operator on two sources, and they are independent vectors. Appendix C shows this decoupling computed on a real dark-energy net.

3 The model

We build the object up from the single agent and then state the population dynamics as an explicit per-round loop (§3.5).

3.1 A paradigm network and its operator

An agent represents a paradigm as a directed network $G = (V, E, \Pi)$: nodes carry theoretical commitments x_v , edges carry inferential dependencies, and each edge bears a coupling precision π_{uv} measuring how tightly v is bound to u . This is just a generative model whose hidden states are the commitments themselves, with precision playing its usual role: a high- π edge says that belief in u strongly constrains belief in v . Internal coherence is the corresponding precision-weighted graphical model, $p(x \mid G) \propto \prod_{(u \rightarrow v) \in E} \psi_{uv}(x_u, x_v)^{\pi_{uv}}$. None of this machinery is ours: it is a linear-Gaussian structural equation model in the lineage of Wright’s path analysis (Wright, 1934), edges as path coefficients and precisions as their weights. We work throughout in that *linear-Gaussian* regime, which puts every quantity below in closed form (Appendix A); agents are *honest*—no one misreports, so lock-in is never a dishonesty artifact—and *bounded*, minimizing variational free energy step by step (Da Costa et al., 2020) rather than planning over policies.

Revising a commitment is not a local act: changing x_v forces everything that depends on it to re-equilibrate. Picture the cascade. A nudge at one node travels along each outgoing edge, weighted by that edge’s precision; from each child it travels on to the grandchildren, picking up their couplings; and so on down every chain of dependence. If A is the matrix of edge precisions, then A^k collects the chains of length k , and summing chains of every length gives $\sum_k A^k = (I - A)^{-1}$ (a finite sum, since a DAG has no cycles to traverse forever). We did not have to invent this object; two literatures already own it. Path analysis knows it as the *total-effects matrix*: what a unit change at one node does to every descendant once all the indirect routes are counted (Wright, 1934; Bollen, 1987). Network science knows the same sum as the Katz–Bonacich index, where it measures how much of the graph a node reaches through walks of all lengths (Katz, 1953; Bonacich, 1987); and it is exactly how influence travels between beliefs in precision-weighted Gaussian models (Malioutov et al., 2006). We adopt it as the *propagation operator* $\mathbf{T} = (I - A)^{-1}$, indexed with rows as sources and columns as descendants, so the entry $\mathbf{T}_{iw} = \pi_{i \rightsquigarrow w}$ is the total precision-weighted coupling propagated from i to w along all directed paths (Appendix A).

Conservatism is then recognized rather than posited. Place a unit of weight on every node and ask, for each commitment i , how much total weight a revision of i would disturb. That is the operator applied to the unit source,

$$\kappa = \mathbf{T} \mathbf{1}, \quad \kappa_i = 1 + \sum_{w \in \text{desc}(i)} \pi_{i \rightsquigarrow w}, \quad (1)$$

the precision-weighted mass of a node’s descendants. In network-science terms this is a centrality score—a node scores high when many strong chains of dependence leave it (formally, the broadcast variant of the Katz index)—and centrality is usually read as influence or status. We read the same number as *revision cost*: a commitment is exactly as expensive to revise as the mass that must re-equilibrate when it moves. Conservatism is thus a centrality the graph already determines, not a disposition assigned by hand.

What we add to this inherited machine comes in two layers. The first is contact with the world. The dense, high-precision interior of the network is the *core*; the sparse periphery in contact with observation is the *belt*, each belt node reading the world through a sensory channel of precision ρ_k . The core carries no direct sensory channel: an interior commitment is revised only by error propagated inward from the belt, through the same operator—costly by Eq. (1) and insulated by position, which is what makes it slow.

The second layer is value, and the operator is run twice. Scientists are bounded agents who value some conclusions over others—a result on which a career rests is held more tightly than its evidence warrants (Kitcher, 1990; Strevens, 2003). Following the variational-cost account of belief

revision (Hyland and Albarracin, 2025), we attach to each node an intrinsic utility u_i and let it propagate through the *same* operator: a commitment one wants true lends value to its dependents, so the conviction field is

$$U = \mathbb{T} u, \quad U_i = u_i + \sum_{w \in \text{desc}(i)} \pi_{i \rightsquigarrow w} u_w. \quad (2)$$

In the operator’s path-analytic reading, U is the total effect of wanting things true. Where the inherited models propagate a single source, a paradigm carries two: $\mathbb{T}\mathbf{1}$ and $\mathbb{T}u$ are linearly independent unless $u \propto \mathbf{1}$, so the two fields need not align (Fig. 1), and Appendix C exhibits the misalignment computed on a real net.

3.2 Motivated revision and the lock-in parameter

The bounded agent revises its belief $q(x)$ over the whole network by trading fidelity to the honest posterior against the value of where the commitments land,

$$\mathcal{F}[q] = D_{\text{KL}}[q(x) \parallel p(x \mid o, G)] - \lambda \mathbb{E}_q[U^\top x], \quad \implies \quad q_\lambda(x) \propto p(x \mid o, G) e^{\lambda U^\top x}, \quad (3)$$

with λ a single temperature governing how far value may bend the evidence. The operation itself is classical—an exponential tilt of the posterior, the form KL-regularized utility maximization always yields (Todorov, 2009; Ortega and Braun, 2013); what is ours is the source of the tilt, a conviction field propagated through the paradigm’s own operator. In the linear-Gaussian regime the tilt is a pure mean shift, displacing the commitments by $\lambda \Sigma U$ and leaving the covariance untouched (Appendix A). What opposes the displacement is evidence, also through the mean: a disconfirming channel k with sensory precision ρ_k and read-out H_k pulls the mean back with gain $\propto \rho_k H_k \Sigma U$. As conviction drives a disconfirming channel toward $\rho_k \rightarrow 0$ that gain vanishes: the channel can no longer move the mean, and the value-displaced commitment stands unrefuted. The fraction of disconfirming sensory precision so silenced,

$$\gamma = \frac{\sum_{k \in \text{dis}} \rho_k \mathbb{K}[\rho_k \text{ gated by the core}]}{\sum_{k \in \text{dis}} \rho_k}, \quad (4)$$

is the *lock-in parameter*. In the experiments γ is endogenous: each round the weight on a disconfirming row is a saturating function of the conviction field projected onto it, $w_k = \exp(-g |U \cdot H_k|)$, with g the conviction-to-censorship coupling—no channel is gated by hand.

3.3 Structure learning: reduction and expansion

Structure learning revises G itself through two asymmetric operations (Friston et al., 2017; Smith et al., 2020), and the asymmetry is the formal counterpart of normal versus revolutionary science.

Reduction is the exact half. Each edge is parameterized by a prior precision α_e ($\alpha_e \rightarrow \infty$ pins the edge’s parameter to zero; $\alpha_e \rightarrow 0$ frees it), the automatic-relevance-determination parameterization of sparse Bayesian learning (MacKay, 1992; Tipping, 2001). Bayesian model reduction (Friston and Penny, 2011; Friston et al., 2018) scores any pruning in closed form from the already-inverted posterior,

$$\frac{p(o \mid \tilde{M})}{p(o \mid M)} = \mathbb{E}_{p(\theta \mid o)} \left[\frac{\tilde{p}(\theta)}{p(\theta)} \right], \quad (5)$$

the Savage–Dickey ratio in general form: one inversion scores every candidate reduced structure without re-touching the data.

Expansion reaches what reduction cannot: a commitment no one has represented. Following Friston et al. (2017), the agent entertains a new node by wiring it tentatively to existing nodes with free priors ($\alpha_e \rightarrow 0$) and letting reduction prune the proposal back to the couplings the data support (Fig. 2). Expansion is not closed-form—a new node may enlarge the likelihood’s support and requires a real inversion—and this cost asymmetry is why a field reorganizes its periphery continually and its centre only rarely. The agent does not plan *when* to expand: candidates arrive at a Poisson rate, and an arrival is kept only if the model log Bayes factor favours the wired-in node (Appendix B). It must be a rate, because one cannot plan toward a dimension one has not yet conceived—Stanford’s problem of unconceived alternatives (Stanford, 2006), to which we return.

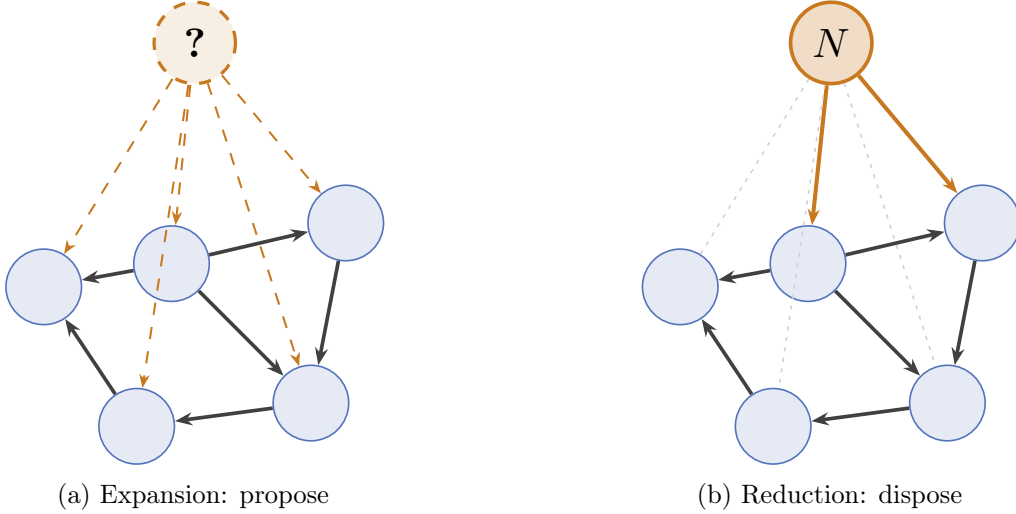


Figure 2: Structure learning as expansion and reduction. a To entertain a commitment not yet in the model, the agent proposes a new node N wired tentatively to every existing commitment with free priors ($\alpha_e \rightarrow 0$, dashed). b Bayesian model reduction prunes the proposal in closed form, keeping only the couplings the data support. Expansion costs a real inversion; pruning is the cheap closed-form half.

3.4 Forgetting

Read once from fixed precisions, $\kappa = \mathbb{T}\mathbf{1}$ can only grow as learning adds edges: a paradigm with no forgetting is a ratchet, and lock-in under it is automatic rather than earned. We import the standard remedy unmodified: accumulated precision relaxes toward the prior between updates, $\Pi \leftarrow \Pi_0 + \omega(\Pi - \Pi_0)$ with forgetting factor $\omega \in (0, 1]$ (Friston et al., 2017b; Smith et al., 2022), so a deposit k steps old carries weight ω^k . The factor is the agent’s prior on the world’s structural drift—one forgets at the rate one believes the structure is moving (Mathys et al., 2011). The operator inherits the time index, $\mathbb{T}_t = (I - A_t)^{-1}$, and a once-central node whose couplings go unreinforced relaxes back toward cheapness. With forgetting, the value tilt saturates at $\lambda U/(1 - \omega)$ and competes with the also-saturated per-step evidence, so conviction holds only above a threshold: an entrenched bloc re-tracks a moving truth at low λ and stays locked at high—motivated persistence turned from a posited bias into a threshold conviction must clear.

3.5 The population, and the simulation at a glance

A population of N agents inhabits a shared hidden world, each carrying its own paradigm network with its own coupling precisions, conviction field $U = \mathbf{T}u$, and derived conservatism $\kappa = \mathbf{T}\mathbf{1}$. Agents are coupled on a trust graph: communities are blocks of a stochastic block model, with dense intra-community edges and an inter-community coupling weight (*inter*) that we sweep from total isolation to dense contact. Beliefs and accumulated precisions are fused across trust edges in information form—a precision-weighted pooling of posteriors in which an agent’s own evidence and its neighbours’ carry weight in proportion to trust, the information-form counterpart of trust-weighted averaging in opinion dynamics (DeGroot, 1974; Acemoglu and Ozdaglar, 2011)—and what is not standard is that structural edits spread the same way: an agent whose neighbour has awakened a new node receives the proposal and scores it against its own model by Eq. (5). Figure 3 draws the picture: what travels on the trust graph is not a number but a piece of structure, evaluated against wiring the sender does not see.

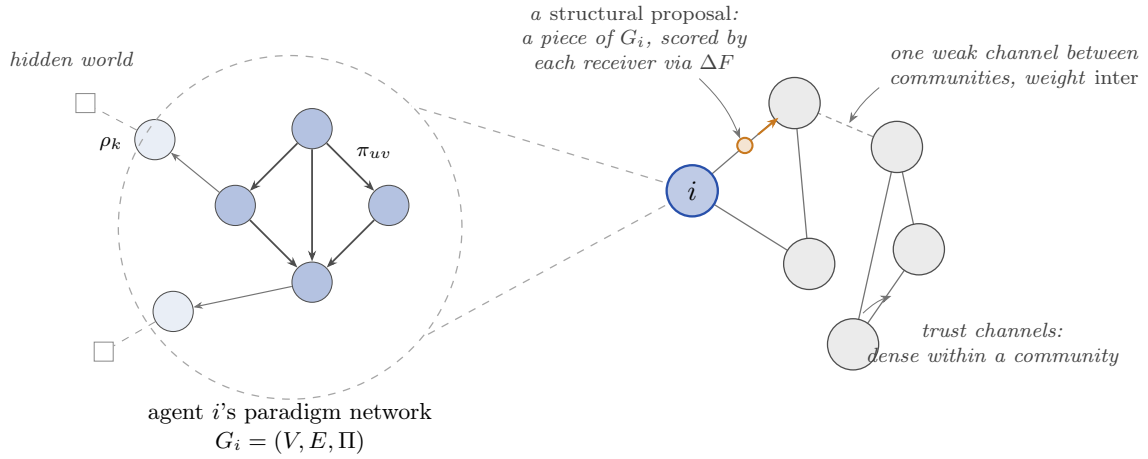


Figure 3: The population model. Right: the community is a trust graph, every circle an agent. Left, magnified: each agent privately holds a paradigm network $G_i = (V, E, \Pi)$ —commitments coupled by precisions π_{uv} , a belt reading the hidden world through sensory channels of precision ρ_k —from which its conservatism $\kappa = \mathbf{T}\mathbf{1}$ and conviction $U = \mathbf{T}u$ are derived. What passes along trust channels (amber) is a *structural proposal*: a piece of paradigm, e.g. a newly awakened node, which each receiver scores against its own network by ΔF (Eq. 5); posteriors fuse precision-weighted along the same channels. The trust graph is fixed at the start (a stochastic block model): channels are dense and strong inside a community, and a single weak weight *inter*—the main dial of §4—connects communities, so what reaches an agent from outside is its neighbours’ evidence and proposals, scaled by trust.

Concretely, the experiments of §4.1 instantiate the phlogiston world: a hub-structured paradigm net in which a single core commitment (phlogiston) is load-bearing for combustion, calcination and respiration; at a fixed step t_{shift} the world’s regime flips (the calx is heavier—the gravimetric anomaly), and the truth thereafter contains a dimension (oxygen) that is *not in any agent’s network*. One round of the simulation, for every agent:

1. **Observe.** The world emits observations at the belt; channel weights on disconfirming rows are set endogenously by the conviction gate, $w_k = \exp(-g |U \cdot H_k|)$ (§3.2), scaled by the agent’s gate strength s .

2. **Infer.** The agent updates its (value-tilted, Eq. 3) posterior over x by exact linear-Gaussian inference and accumulates precision.
3. **Crisis check (reduction).** Each candidate prune is scored on the move ledger: the closed-form prune evidence ΔF (Eq. 5) plus λ_i times the equally closed-form conviction-value change of the prune, $\Delta U = U^\top(\mu_{\text{reduced}} - \mu)$, read from the reduced posterior the Savage–Dickey identity already exhibits. Both protective belt edges scoring $\Delta F + \lambda_i \Delta U > 0$ is a *crisis*, and the agent applies the reduction—the hub is released (Appendix B).
4. **Discovery check (expansion).** After reduction, the residual the old paradigm absorbed becomes visible as a coherent pattern in prediction error; when a Poisson proposal arrives (rate r), its *direction* is read from that pattern’s leading eigenvector, and the wake is accepted iff the model log Bayes factor is positive—the data must hold the new node (§3.3, Appendix B).
5. **Fuse.** Posteriors and structural edits are pooled across the trust graph, weighted by the intra/inter coupling.
6. **Forget.** Accumulated precision decays by ω (§3.4).

Crisis, revolution and discovery are thus not stipulated events but threshold crossings of the model’s own evidence; Appendix B derives both checks from the single tilted objective and lists explicitly what remains a modelling commitment. Table 1 lists the dials swept; everything else is held fixed, and all code is seeded and ships its own null and positive controls.

Dial	Symbol	What it controls
Social coupling	<i>inter</i>	inter-community trust weight (0 = isolation)
Vanguard fraction	—	fraction of agents in the open (low- λ) community
Conviction	λ_d	dogmatic community’s value tilt / prune protection
Gate scaling	s	strength of the conviction gate on disconfirming channels
Proposal rate	r (Poisson)	arrival rate of structural proposals
Forgetting	$\omega \in (0, 1]$	decay of accumulated evidence toward the prior
Observation noise	σ_o	sensory noise at the belt

Table 1: The dials swept in the experiments (1,816 runs); all other quantities are held fixed.

4 Results

All results are from the population of §3.5 run through the full Kuhnian cycle on the phlogiston world, swept overnight over the dials of Table 1 (1,816 healthy runs). The single-agent *dark-energy* net that realises the two fields of Fig. 1 on a worked example—and recovers the Ω_m – Λ degeneracy as a Schur complement, a quantity a scalar credence has no room to express—is reported in Appendix C. The nets are hand-wired to expose the geometry, not fitted to data: nothing here is evidence about chemistry or cosmology, only about how a structured, bounded, valued mind changes. We close with limitations and future work (§4.2).

4.1 The full cycle, endogenously: the phlogiston sweep

In the population, the whole Kuhnian arc—normal science, the gravimetric anomaly, crisis, reduction, discovery of oxygen, new normal—unfolds as successive threshold crossings of the agents’ own evidence (§3.5), with no phase stipulated. The sweep then asks where each beat of the arc lives in the dial space, and three findings organize it.

Even a trace of outside contact ends the old paradigm; only total isolation protects it. The first question the sweep answers: how much contact with the rest of the world can a dogmatic community tolerate and still keep phlogiston? Almost none. The setup pits a dogmatic community—high conviction, which makes its agents gate their own disconfirming channels (§3.2)—against an open vanguard already tracking the anomaly, and dials the trust weight between the two communities (*inter*) from 0 (sealed off) up to dense contact. Sealed off, the protection works: under strong gating ($s = 10$) only 19% of the dogmatic community ever converts to oxygen. But the seal has to be perfect. At *inter* = 0.005—trust toward outsiders two hundred times weaker than trust within the community—92% convert, no matter how strong the gate, and making the agents twice as dogmatic ($\lambda_d : 0.35 \rightarrow 0.8$) does not shift the picture at all (Fig. 4). The transition is genuinely a boundary, not a gradual fade: the run-to-run spread of outcomes, which is large exactly where a run can tip either way, spikes in a narrow band around *inter* $\approx 10^{-3}$ and is flat on both sides. Why is contact so corrosive? Because fusion shares *evidence*, not just conclusions: an open vanguard of just 5% of the population lifts end-state conversion from 0.60 to 0.69, because the vanguard’s anomalous data flows across the trust channels and sets off each dogmatist’s *own* crisis. An agent can silence its own instruments; it cannot silence its neighbours’. Gating cannot defeat connection.

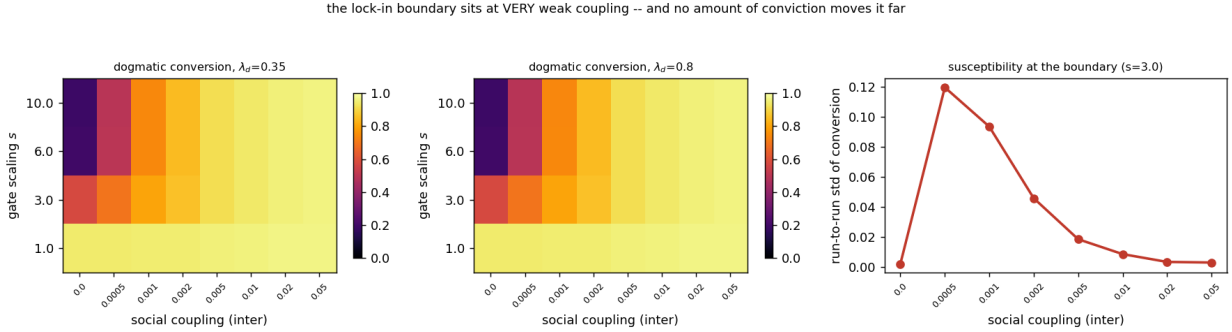


Figure 4: Only isolation protects a superseded paradigm (phlogiston sweep). *Left, centre:* the fraction of the dogmatic community that converts to oxygen, as a function of contact with the open community (*inter*) and of how strongly conviction gates disconfirming channels (s), at two conviction levels ($\lambda_d = 0.35$ and 0.8). Gating keeps conversion low only when contact is essentially zero; past *inter* ≈ 0.005 nearly everyone converts, and doubling conviction barely changes the picture. *Right:* the run-to-run spread of outcomes—large exactly where a run can tip either way—spikes in a narrow band around *inter* $\approx 5 \times 10^{-4}$, the signature of a sharp boundary rather than a gradual fade.

A control pins down what does the stalling. Keep conviction exactly as high but switch the gate off—the agents stay just as motivated, but keep listening—and the community revolts in full (revolted fraction 1.0, versus 0.0 once the gate silences the disconfirming channels, $\gamma \rightarrow 1$). So the evidence was never absent or out of reach: what stalls a community is conviction closing the channels that would deliver it. Lock-in is derived, not assumed—it appears exactly when enough disconfirming precision is silenced (Eq. 4) that the data lose their grip on the agent’s commitments.

Discovery waits on evidence; keeping it requires discovering together. Discovery in this model means waking a node—oxygen—that no agent’s network yet contains, and since one cannot search for a concept one has not conceived, candidates arrive at random (Poisson rate r , §3.3). Individually this behaves as a waiting process should: the fraction of the open community that discovers within the horizon follows the textbook law $1 - (1 - r)^T$, rising from 44% at $r = 0.01$ to 100% at $r = 0.16$ (Fig. 5, left). The *first* discovery, however, is not rate-limited at all: with ~ 40

agents drawing in parallel, an attempt arrives almost immediately at every rate, and the median first discovery sits at $t \approx 82$ whether proposals are rare, frequent, or deterministic (right panel)—its date is set by when the post-reduction residual earns a positive Bayes factor, not by the arrival process. The community’s first discovery waits on evidence, not on luck. What the rate *does* govern is survival. A discovery made by one agent at a time does not survive (the red curve): at $r \leq 0.04$ the lone discoverer’s oxygen node is averaged away every time—the moment it fuses beliefs with peers who lack the concept, the shared prior treats the slot the others do not represent as a firmly held “no such thing,” and pulls the awakened coupling back to zero. Survival appears only once arrivals are frequent enough to wake the community within a few steps of one another: 31% of runs at $r = 0.16$, matching the simultaneous-proposal limit (28%). A new concept does not need a critical mass of believers eventually; it needs them at the same *time*. (On the fusion scheme this result rides on, see §4.2.)

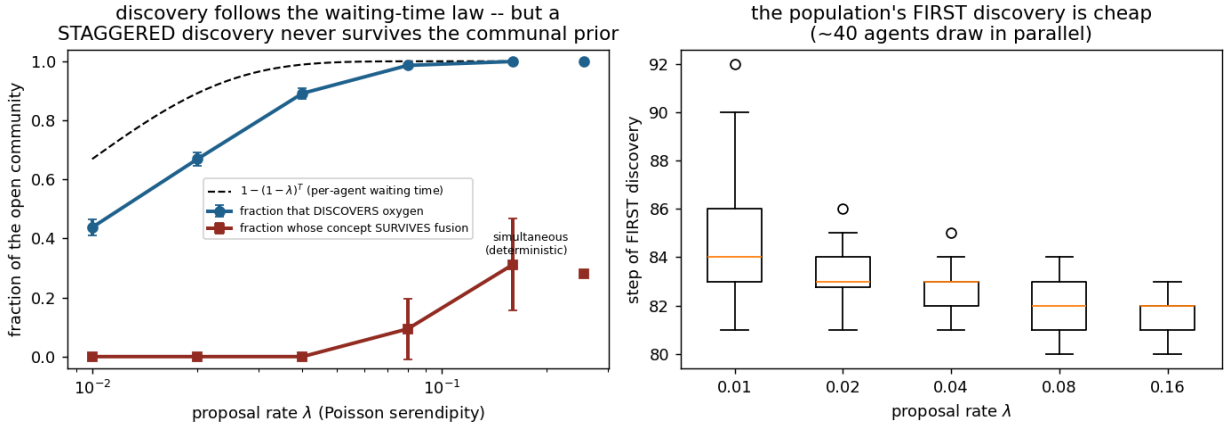


Figure 5: Discovery and survival of the unconceived (phlogiston sweep). *Left*: the fraction of the open community that discovers oxygen follows the waiting-time law $1 - (1 - r)^T$ (dashed); the fraction whose concept *survives* fusion (red) is zero at slow rates—staggered discoverers are pulled back by the communal prior—and rises only once arrivals cluster the wakings in time, reaching 31% at $r = 0.16$, the simultaneous-proposal limit (28%, point at right edge)—critical mass in *time*. *Right*: with ~ 40 agents drawing in parallel, an attempt arrives almost immediately at every rate; the date of the first discovery is set by the evidence (the post-reduction Bayes factor), not by the arrival process.

The cycle is robust, and the discovery check carries no tuned threshold. Re-running the full arc across forgetting and observation noise ($\omega \in [0.9, 1.0]$, $\sigma_o \in [0.25, 1.0]$) leaves crisis and reduction untouched everywhere—including the no-forgetting wall $\omega = 1$ —and discovery essentially complete at every $\omega < 1$. At the wall itself discovery thins (55%/43%/16% of the open community at $\sigma_o = 0.25/0.5/1.0$): with nothing ever forgotten, the accumulated precision is so entrenched that the woken node struggles to earn a positive Bayes factor against it. A community that never forgets still sees its paradigm die; what it loses is the capacity to grow the successor. Nor is there an acceptance bar to miscalibrate: the wake’s sole accept test is the model log Bayes factor, which rejects proposals the residual does not support (Appendix B). A separate residual-floor gate—demanding the unexplained error clear some height before a new node is even entertained—would duplicate that test with a free constant, and a sensitivity probe shows the constant is dangerous: set

above the prediction error the dead paradigm actually leaves behind after reduction, it suppresses every proposal and discovery dies almost everywhere. The model instead has the property that gate would have had to be tuned to imitate: an agent entertains new structure at exactly the level of error its old structure produces, because that error is what the Bayes factor weighs.

4.2 Limitations and future work

The honest boundary of the model is easy to draw because its two threshold events are derived: crisis and discovery are the move ledger of Appendix B, and whatever the derivation does not fix is named there as a commitment.

Model expansion is the narrow waist. Everything revolutionary in the model passes through one move: waking a single, pre-allocated slot (oxygen), wired to a fixed candidate set, in the direction of the leading eigenvector of a windowed residual. That is the thinnest possible model of conceptual innovation. The candidate space is not searched, it is given; a real discovery process entertains many structures at once—several hubs, new couplings among existing commitments, new observation channels—and the rank-one reading of the residual is exact only for a single hidden common cause. The acceptance rule is equally minimal: the wake is kept iff the model log Bayes factor is positive—curiosity (the epistemic gain ΔG) pays for *attempting* the inversion, evidence for *keeping* the result—which collapses into one step what is more naturally a wake-then-prune cycle in which a community entertains structures on credit before the data settle them (§3.3). And the arrival of candidates is a constant-rate Poisson process—the maximally agnostic stand-in for a generative process the model does not have, deliberately so (one cannot plan toward an unconceived dimension), but a stand-in nonetheless. Appendix B.5 sketches the two refinements we consider most promising: a mutually-exciting (Hawkes-type) proposal rate riding the trust graph, the mechanism by which a community could manufacture the near-simultaneity that concept survival requires; and the rate as an updatable hyperprior on the agent’s own model adequacy, driven by the residual the discovery check already records.

What remains stipulated. Appendix B enumerates what the derivation does not fix—the operationalization of crisis, the ordering of crisis before expansion, the estimator horizons. One commitment deserves the main text: the conviction source u is exogenous and static. Letting u accrete onto structurally entrenched commitments—value aligning with conservatism as a programme matures—would recover Lakatos’s progressive and degenerating problemshifts and meet Laudan’s reticulated co-variation of theory, method, and aim (Lakatos, 1970; Laudan, 1984).

Fusion decides what survives, and fusion is a modelling choice. The staggered-discovery death of Fig. 5 rides on posterior fusion pooling the unconceived slot’s pin as if it were a firmly held “no such thing.” That reads naturally as incommensurability—a shared conceptual scheme suppressing concepts it does not contain—but fusion schemes that exclude unrepresented dimensions would soften the effect, and the survival threshold for a new concept would move with them. Which pooling rule real communities implement is an empirical question this model can only pose.

The staircase lives in observability, and what that localizes. The structural account predicts a belt-first, core-last *staircase*: the cheap periphery reorganizes first, the costly core only after the crisis. Where the staircase comes from turns out to be sharply diagnostic. A conservatism-gated revision *rate* does not produce it—the high- κ nodes reorganize first, indistinguishably from uniform

and shuffled- κ nulls (Appendix F)—because the linear-Gaussian Fisher deposit is set by the observation operator rather than the data, so the conservatism geometry never enters the update. The staircase appears when that geometry is placed in *observability*: when core commitments carry no direct observation rows and evidence reaches them only through relational channels, the belt realigns immediately after the world changes and the core only after the agent’s own crisis, while the full-observability configuration—no staircase—serves as the control. What this localizes as the live limitation is the operator-set deposit itself: couplings here are selected among, not estimated from data, so structure learning is model comparison over a parameterized family rather than discovery of arbitrary structure. Each experiment ships its own null and positive controls (a world that never shifts produces neither crisis nor discovery; the gated, isolated community stays locked; Poisson discovery follows the waiting-time law), and the provenance of the sweep numbers is flagged where they are quoted (§4.1).

Finally, the model is stylized. The nets are hand-wired to expose the geometry, not fitted to data; the dials of Table 1 are swept, not inferred; and a community that “tracks” the eventually-true theory is committed to it, not thereby correct. What clears the “more than a redescription” bar is not any single result but the conjunction a scalar credence cannot voice: the Schur-complement degeneracy (a scalar has no off-diagonal to fill), the derived-not-imposed lock-in with its deaf-but-honest control, and an expansion that wakes an unconceived node only when the world makes its absence felt—now with no tuned threshold anywhere between the evidence and the act.

5 Discussion

Everything distinctive follows from taking the hidden structure seriously: a conservatism gradient derived rather than assigned, an exploration drive that is the value of expanding the model rather than a tuned bonus, a conviction field that is the same operator on a different source, and a population arc—crisis, revolution, lock-in, or release—that emerges as threshold crossings of the agents’ own evidence. The sweep adds the population-scale phenomenology: the lock-in boundary lives at very weak coupling, so the protective strategies internal to a community (conviction, gating) are dominated by the single external variable of contact; and the survival of a new concept is governed not by how many eventually conceive it but by how nearly in time they do.

We should be candid about which philosopher this picture serves: Lakatos, more than Kuhn. The conservatism geometry *derives* his hard core and protective belt—a node is costly to revise because much depends on it—rather than decreeing them (Lakatos, 1970). Identifying the carry-over cascade with Kuhnian *incommensurability* is the framing’s most contestable move: a directed Bayes net makes incommensurability commensurable by construction—one precision space, revolution a long traverse rather than a gestalt switch—whereas Kuhn’s incommensurability is precisely the absence of the common measure the network presupposes (Kuhn, 1983). Where the staircase lives (§4.2) is the tell: it appears only when the core is shielded from direct observation, and graded reorganization on fixed, fully observed wiring is a Lakatosian problemshift, not a Kuhnian reconstitution. Yet the sweep’s survival result earns a piece of it back: the fusion that crushes a staggered discovery is a shared conceptual scheme suppressing a concept it does not contain—incommensurability as a population-level force, not a stipulated incomparability.

Finally, the unconceived marks the limit of structure learning itself. Structure learning navigates beautifully within the dimensions a model posits but is blind to the dimension no one has proposed—Stanford’s problem of unconceived alternatives (Stanford, 2006): not uncertainty over which known structure is right, but whether the space of known structures contains the truth at all. One cannot

place a posterior over the unconceived; running an expansion process that keeps proposing nodes at some Poisson rate is what a healthy epistemic system does instead of pretending it can. §4.1 shows both edges of that blade: too low a rate leaves the dimension undiscovered within the horizon, and a discovery made alone does not survive the communal prior. The hidden structure of the world, which forced us into structure learning on the first page, is in the end the thing no amount of structure learning can guarantee to have caught.

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A The operator, the tilt, and the Schur residue

Linear-Gaussian construction. The couplings form a structural equation model $x = Ax + \zeta$, $\zeta \sim \mathcal{N}(0, D^{-1})$ with A the weighted DAG adjacency (Koller and Friedman, 2009); the choice is for tractability, and a nonlinear coupling would replace T by a local linearization. The propagation operator is the resolvent

$$\mathsf{T} = (I - A)^{-1} = \sum_{k \geq 0} A^k, \quad \mathsf{T}_{iw} = \sum_{\text{paths } i \rightsquigarrow w} \prod \pi = \pi_{i \rightsquigarrow w}, \quad (6)$$

the Neumann series finite because a DAG makes A nilpotent. We realise T on the coupling magnitudes $|A|$ so downstream mass does not cancel when a child opposes a parent; the signed A builds the precision matrix. We index T so that κ_i sums row i ; under the SEM convention the released code stores the transpose and computes the same fields as column-sums of $\mathsf{T}^\top$, so a reader reproducing **T1** literally against the code’s array must transpose first.

The value tilt. With honest posterior $p(x \mid o, G) = \mathcal{N}(\mu, \Sigma)$, the tilted posterior of Eq. (3) is a pure mean shift, $q_\lambda(x) = \mathcal{N}(\mu + \lambda \Sigma U, \Sigma)$, with cumulant-generating function $Z(\lambda) = \lambda U^\top \mu + \frac{1}{2} \lambda^2 U^\top \Sigma U$. Note that gating a channel ($\rho_k \rightarrow 0$) *raises* the posterior variance and hence Z'' ; lock-in lives in the vanishing mean-update gain $\rho_k H_k \Sigma U$, not in the curvature.

The Schur residue. Marginalizing a hub condenses its influence into new edges among its neighbours: for a common-cause structure over (x_1, x_2, b) ,

$$\Pi = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix} \xrightarrow{\text{marginalize } b} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \end{pmatrix} = \begin{pmatrix} 1.5 & -0.5 \\ -0.5 & 1.5 \end{pmatrix}, \quad (7)$$

a direct coupling induced where the joint had a zero. Conditioning on a shared *effect* (the supernova collider of Appendix C) is the dual operation, and both fill in the off-diagonal coupling the carry-over cost is built from.

B The crisis and discovery checks, derived from the ledger

The per-round loop of §3.5 contains two threshold events: the *crisis check* (when does an agent prune the protective edges and release the hub?) and the *discovery check* (when does it wake the unconceived node?). This appendix states exactly what each check computes, derives both from the single objective the paper already carries, and lists the modelling commitments that remain after the derivation. The aim is that nothing in the two checks should read as a stipulated event: every acceptance is the model’s own evidence weighed on the model’s own ledger, and everything that is *not* derived is named as a commitment.

B.1 One ledger for every structural move

The bounded agent of §3.2 revises its beliefs by minimizing $D_{\text{KL}}[q \parallel p(x \mid o, G)] - \lambda \mathbb{E}_q[U^\top x]$ (Eq. 3): fidelity to the honest posterior, traded against the value of where the commitments land, at temperature λ . A structural move—pruning an edge, waking a node—changes both terms, so the same objective scores the move:

$$\text{score}(M \rightarrow \tilde{M}) = \underbrace{\Delta F}_{\text{log Bayes factor}} + \underbrace{\Delta G}_{\text{epistemic gain}} + \lambda \underbrace{\Delta U}_{\text{conviction value}}, \quad \text{accept} \iff \text{score} > 0, \quad (8)$$

with ΔF the evidence for the edited structure, ΔG the expected information gain about any newly introduced latent (zero for a prune), and $\Delta U = \mathbb{E}_{q_{\tilde{M}}}[U^\top x] - \mathbb{E}_{q_M}[U^\top x]$ the change in the conviction-field value of the commitments. Equation (8) is not a new mechanism: it is Eq. (3) evaluated at two structures and differenced. Both threshold events of the simulation are instances of it.

B.2 The crisis check is the ledger on a prune

For a prune, every term of Eq. (8) is closed-form in the linear-Gaussian setting. The evidence half is Bayesian model reduction (Eq. 5): writing the agent’s accumulated likelihood deposit as $J = \Pi - \Pi_0$, $j = h - h_0$, the Savage–Dickey identity scores the swap of the prior (Π_0, h_0) for the edge- e -reduced prior $(\Pi_0^{(e)}, h_0^{(e)})$ under the same likelihood, and—this is the part the main text does not spell out—it *already exhibits the reduced posterior*,

$$\Pi^{(e)} = \Pi_0^{(e)} + J, \quad h^{(e)} = h_0^{(e)} + j, \quad (9)$$

so the value half costs one linear solve per candidate edge:

$$\Delta U_e = U^\top (\mu^{(e)} - \mu), \quad \mu^{(e)} = (\Pi^{(e)})^{-1} h^{(e)}, \quad \mu = \Pi^{-1} h, \quad (10)$$

with $U = \mathbf{T}u$ the agent’s *current* conviction field, read along its present couplings. The crisis check is then exactly the ledger: edge e is flagged iff $\Delta F_e + \lambda \Delta U_e > 0$, and *crisis* is the round at which both protective belt edges are flagged, upon which the agent adopts the reduced prior composed over every edge its own ledger flags (the hub released). Pruning a conviction-protective edge moves the posterior mean away from the cherished commitments, so $\Delta U_e < 0$ and the rule reproduces the intuitive form “the evidence must overcome λ times the value the edge protects.”

A natural surrogate suggests itself here: the static threshold $\Delta F_e > \lambda v_e$ with $v_e = |U_{\text{parent}}| + |U_{\text{child}}|$, the summed conviction magnitude at the edge’s endpoints—available without a solve, and with the right *form* (it is Eq. (8) with $\Delta U_e \equiv -v_e$). But it has the wrong *content*: v_e is a static stand-in for a quantity the model can compute exactly, so we compute it. The difference is not cosmetic, and its direction is itself informative: because the reduced and full posteriors share the likelihood deposit, the mean displacement of a prune—and with it $\lambda \Delta U_e$ —*shrinks as evidence accumulates*. A data-rich paradigm is barely protected at the threshold; almost all of its persistence must come from upstream, from the conviction gate that decides which evidence is allowed to accumulate at all (Eq. 4). The static surrogate would quietly overstate threshold protection; the exact ledger locates lock-in where the paper’s results put it: gating plus isolation, not a high bar at the moment of decision.

B.3 The discovery check is the ledger on an expansion

Expansion cannot be closed-form (§3.3): a new node needs a real inversion, and—Stanford’s problem of unconceived alternatives—the agent cannot plan toward a dimension it has not conceived. The discovery check therefore has exactly two parts, and only two:

1. **Arrival.** Candidate proposals arrive at a Poisson rate r . This is the one place a stipulated process enters, and it is a deliberate modelling commitment: a rate is the maximally agnostic model of conceiving what one has not conceived. The proposal’s *direction* is read from the model’s own prediction errors: a hidden hub b wired to the visible nodes with couplings c leaves, on marginalization, the rank-one footprint $-cc^\top / \Pi_{bb}$, so the leading eigenvector of the windowed error second moment is the best single-hub explanation of the residual. The window is an estimator horizon, not a gate.

2. **Acceptance.** The arrived proposal is accepted iff the *model log Bayes factor* of the bordered (hub-wired) model against the agent’s current one is positive—that is, iff the residual genuinely loads on the proposed hub, equivalently iff the immediate Savage–Dickey prune of the new couplings would be rejected. The epistemic gain ΔG sits on the ledger too, but on the other side of the move: being positive for *any* new latent, it is what pays for attempting the (in general expensive) inversion, and it cannot keep a hub the data do not hold—an accept test that included it would wake on pure noise. Entertaining is paid by curiosity; keeping is paid by evidence. One consequence is owned rather than hidden: repeated looks give the Bayes factor a small false-discovery rate (rare noise crossings of $\Delta F > 0$ in the null world). We regard this as a feature of the honest model—communities do occasionally adopt spurious structure—and its principled remedy is not a reinstated threshold but the wake-then-prune cycle named in the limitations, in which a spuriously woken node is pruned back by the same reduction that scored it.

A tempting third part is deliberately absent: a *trigger*, requiring the residual floor (the eigenvalue above) to clear some height, perhaps for a sustained number of steps, before the Bayes factor is consulted. Such a gate would duplicate the accept test’s job with a free constant—the Bayes factor already rejects proposals the residual does not support—and its only principled reading, a budget on when to pay for the expensive inversion, is a statement about real scientific communities rather than about this simulation, where the inversion is a small log-determinant. The discovery check therefore has no tuned threshold: the Poisson rate carries Stanford’s commitment, the eigenvector carries the direction, and the ledger decides. (The floor is still recorded as telemetry, because it is the visible signature of the residual the old paradigm had been absorbing.)

B.4 What remains a modelling commitment

The derivation above leaves a short, explicit list of choices that are *not* forced by the formalism. We state them because the paper’s claim that “crisis, revolution and discovery are threshold crossings of the model’s own evidence” is honest only if its boundary is drawn:

- **The belt conjunction.** “Crisis” is operationalized as *both* named mass-law edges flagged in the same round. The per-edge accept is derived; the conjunction over those two edges is our operational definition of having eliminated the phlogiston mass-law reading.
- **The ordering.** Expansion is only attempted strictly after an agent’s own crisis. The narrative justification (the residual the old paradigm absorbed becomes visible only after reduction) is mechanistically plausible but enforced rather than emergent; letting the check run pre-crisis and showing the ledger rejects it would upgrade this commitment to a result.
- **The Poisson rate r** (swept), the prediction-error *window* (an estimator horizon), a short *warm-up* before reduction is permitted (an initialization guard), and the released hub’s self-precision are quantitative choices of the same standing as the other dials of Table 1.
- **The conviction field in the ledger.** ΔU is evaluated under the agent’s current conviction field $U = \mathbb{T}u$ with the intrinsic utility u fixed; agents do not revise u itself. Motivated agents who re-derive what they value is a different (interesting) model.

B.5 Future work: the rate, made principled

Exploration as a social process. The Poisson proposal rate is deliberately the most agnostic model of conceiving an unconceived alternative, and it is also where the model is most clearly

incomplete: in real communities the *propensity* to entertain new structure is itself socially modulated. A natural refinement keeps Stanford’s constraint (no agent can plan toward a specific unconceived dimension) while letting the rate ride the trust graph: a self- and mutually-exciting (Hawkes) process, in which agent i ’s rate $r_i(t)$ receives a decaying bump whenever a trusted neighbour expands,

$$r_i(t) = r_0 + \sum_{j \in \partial i} T_{ij} \sum_{t_j < t} \beta e^{-(t-t_j)/\tau}, \quad (11)$$

with t_j the neighbours’ discovery times. This is the structural-proposal channel the model already has, promoted from carrying *content* (which node to wake) to also carrying *momentum* (how eagerly to look)—a formal home for the observation that revolutions arrive in waves. Because the population results show a staggered discovery never survives the communal prior, an excitation term is not a cosmetic addition: it is the mechanism by which a community could manufacture the near-simultaneity that survival requires, and the Hawkes branching ratio would become the model’s answer to how much social excitement a revolution needs.

The rate as a belief: an active-inference closure. A second refinement folds the rate back into the formalism. The forgetting factor is already read as a prior on structural drift (§3.4: one forgets at the rate one believes the world is moving), and the proposal rate admits the dual reading: one *explores* at the rate one believes one’s model is incomplete. That belief is itself updatable—a hyperprior on model adequacy, driven by the very residual floor the discovery check records as telemetry—so that a persistent unexplained residual raises the agent’s own arrival rate, while a quiet model lets it decay. Under that closure the exploration rate stops being a dial and becomes a posterior; together with the epistemic-gain term already on the ledger, expansion would then be fully inside the active-inference loop (the agent expands when expected information gain pays for the inversion, and looks for candidates at a rate set by its inferred inadequacy). We did not pursue this here because it dissolves the clean waiting-time law into a feedback system, and because the general problem it gestures at—principled out-of-distribution generalization, how an agent should behave toward dimensions it cannot yet represent—is well beyond this paper’s scope. The phlogiston world needs only the agnostic rate; the refinement is where this model meets that open problem.

C The two fields on a worked example: the dark-energy net

This appendix realises the two fields of Fig. 1 on a domain net: general relativity (GR) at the root; the density parameters Ω_m and the cosmological constant Λ as dials; the Friedmann expansion, distance modulus $\mu(z)$ and a Type-Ia supernova (SN) reading as a descending chain; light-bending, the perihelion precession, gravitational redshift, CMB and BBN as GR’s other branches; and a modified-gravity (MOND) prediction the community wants false. The net is hand-wired to expose the geometry, not fitted to data.

Computing $\kappa = \mathbf{T}\mathbf{1}$ and $U = \mathbf{T}u$ on this net (Fig. 6) makes the two-field claim concrete. Conservatism is steepest at GR ($\kappa_{\text{GR}} = 10.5$, the maximum) and shallow at the cosmological constant ($\kappa_{\Lambda} = 2.85$, a ratio of 3.7): GR carries every predicted observable below it, Λ only the distance–redshift branch. This is Lakatos’s belt-protects-core falling straight out of the operator—when the 1998 anomaly arrives at the bottom, the cheap move is to turn the dial with the smallest descendant set (Λ), not the ancestor with the largest (GR). The conviction field $U = \mathbf{T}u$, with the community valuing the accelerating-universe supernova reading ($u_{\text{SN}} = +1$), the Λ commitment (+0.6) and wanting MOND false (−0.8), *inverts* that order: $U_{\Lambda} = 1.19$ (the most cherished) exceeds $U_{\text{GR}} = 0.45$ while $\kappa_{\Lambda} < \kappa_{\text{GR}}$. The two fields are de-correlated (Pearson $r(\kappa, U) = 0.25$), the

visible face of the algebraic independence guaranteed whenever $u \not\propto \mathbf{1}$: Λ is *cheap yet cherished*, GR *central yet comparatively value-neutral*, the GR machinery (light-bending, perihelion, clocks, CMB, BBN) central yet value-neutral ($U \approx 0$), and MOND the conclusion the community wants false ($U < 0$)—exactly the three annotated cases of Fig. 1, now computed by the operator on a domain net rather than assigned by hand.

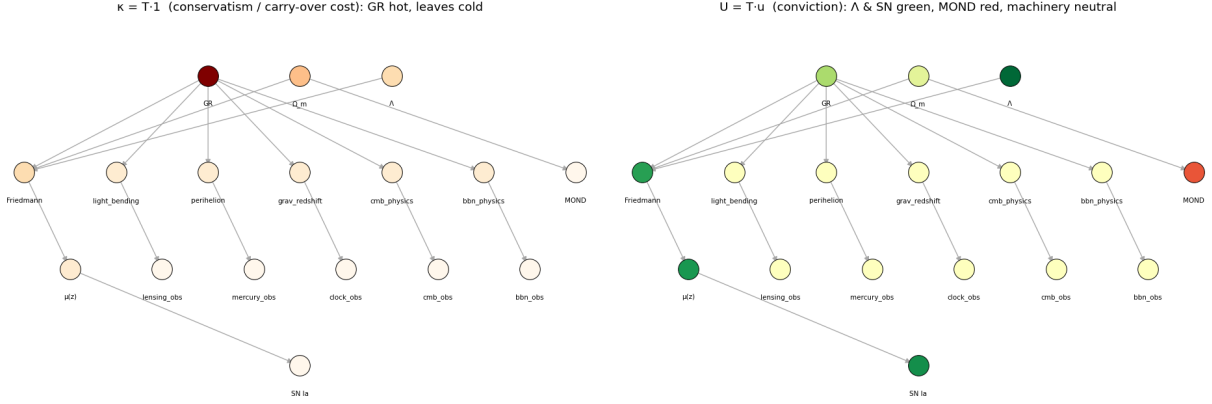


Figure 6: The two fields, computed on the dark-energy paradigm net. *Left*: conservatism $\kappa = T\mathbf{1}$ (Eq. 1)—GR is hot ($\kappa = 10.5$), the observation leaves cold; the directed arrows make “everything hangs below GR” visible. *Right*: conviction $U = T\mathbf{u}$ (Eq. 2) on the same graph— Λ and the supernova reading green (cherished), MOND red (wanted false), GR’s other machinery value-neutral. The decoupling is the realisation of Fig. 1: the value order inverts the cost order ($U_\Lambda > U_{GR}$ while $\kappa_\Lambda < \kappa_{GR}$; $r(\kappa, U) = 0.25$).

The Schur residue of Eq. (7) is equally concrete. A priori Ω_m and Λ are independent (correlation 0.00, a round cloud). Conditioning on the shared supernova observation—clamping the descendant both dials feed—manufactures an off-diagonal coupling between them where the joint had a zero: the cloud collapses onto the tilted “banana” (Fig. 7), with correlation rising to +0.50. This is the Ω_m – Λ degeneracy direction every dark-energy contour plot draws, here reproduced as a Schur complement on the precision matrix. Conditioning on a shared *effect* (the supernova collider—explaining-away) and marginalizing a shared *cause* (Eq. 7) are *dual* operations, not the same one, but both fill in the off-diagonal coupling the carry-over cost is built from.

D Coupling of the two fields under model expansion

The two fields are coupled by the expansion operation, and with T defined the coupling follows directly. When the agent expands its model—proposing a new commitment N wired by candidate conditionals to a set of existing nodes—the joint gains an explicit factor,

$$p(x, x_N | G') \propto p(x | G) \prod_{(N \leftrightarrow v)} \psi_{Nv}(x_v, x_N)^{\pi_{Nv}}, \quad (12)$$

with the candidate edges entered at free prior precision ($\alpha_e \rightarrow 0$) so that reduction can carve them back. In operator terms, adding N appends a row and a column to A —and therefore to T —and appends one entry to *each* of the two source vectors, the unit source $\mathbf{1}$ and the utility source u . The

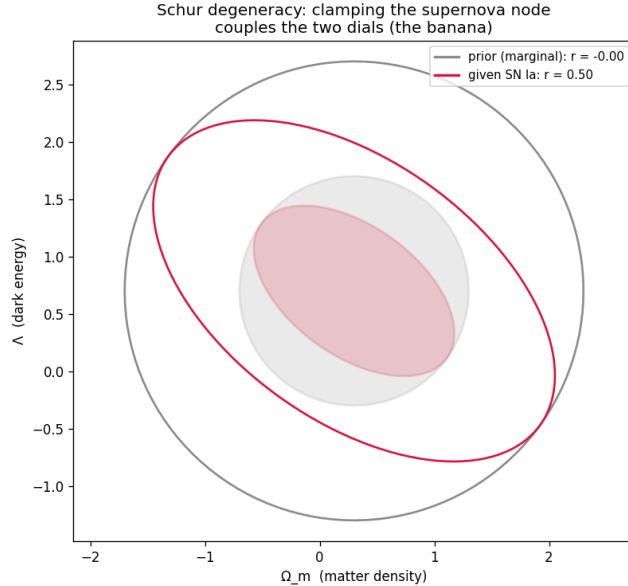


Figure 7: The Schur degeneracy, computed. The two dials are uncorrelated a priori (grey, $r = 0.00$); conditioning on their shared supernova descendant induces the off-diagonal coupling (crimson, $r = +0.50$)—the Ω_m - Λ “banana”. Marginalizing a shared latent hub and conditioning on a shared observation are the two readings of one complement.

two consequences are then immediate. Every node N touches gains the new coupling precision it must now also propagate,

$$\Delta\kappa_v = \pi_{Nv}, \quad (13)$$

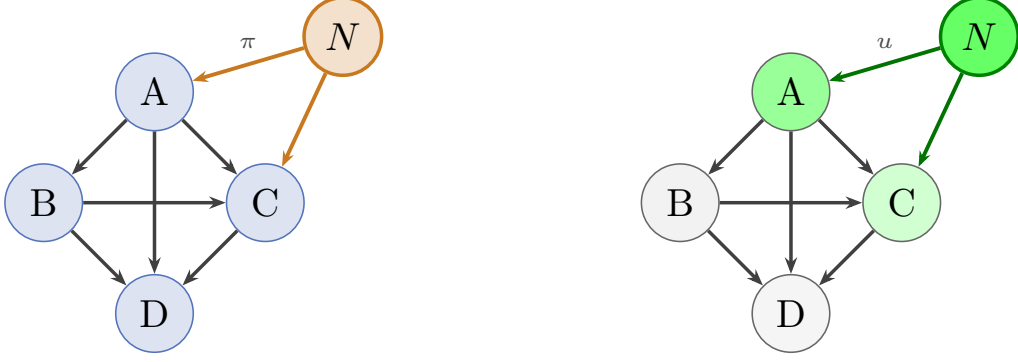
while the conviction field gains, at N and along its new edges,

$$\Delta U = u_N + \sum_{w \in \text{desc}(N)} \pi_{N \sim w} u_w. \quad (14)$$

A single expansion thus writes the new commitment into both the cost geometry and the value geometry (Fig. 8). This is the precise content of motivated structure learning: an agent contemplating a node that would, once wired in, lend value to a cherished core—or force the revision of one—registers that prospect in *both* fields, as a carry-over cost it must pay and as a value change it may welcome or resist, and scores the keep-or-prune decision against the value-tilted ledger of Eq. (3). The same proposal can therefore be evidence-favoured and value-rejected, or the reverse.

E A coarse model resolves a complex world only slowly

This experiment isolates the cost of a coarse Bayes net against a world richer than it. The true world is a complex twelve-parameter Gaussian with four latent factors (correlated, individually ambiguous directions); a coarse agent senses only res random aggregate combinations per step. Coarser is slower (Fig. 9, left): the time to resolve the world state to within a fixed tolerance grows from 58 steps at full resolution to 221 steps at $\text{res} = 1$, and below a resolution the agent does not finish within the horizon. Adding forgetting floors the resolution outright (Fig. 9, right): a net that both under-senses and forgets plateaus at a residual error that rises as ω falls ($0.08 \rightarrow 0.28$), and at $\text{res} = 3$, $\omega \leq 0.93$



(a) enters the belief network (coupling π , $\Delta\kappa_v = \pi_{Nv}$) (b) enters the utility network (propagated value, ΔU)

Figure 8: One expansion, two fields. Adding a commitment N with new conditionals appends a row and a column to the operator $\mathbf{T}$ and an entry to each source vector: into the belief network, where each new edge carries a coupling precision π and so adds to carry-over cost by Eq. (13) (a), and into the utility network, where N 's value propagates along the same new edges by Eq. (14) and tints the commitments it reaches (b). The keep-or-prune decision is scored against the value-tilted ledger, so a proposal can be evidence-favoured yet value-rejected, or the reverse.

never resolves the world at all. A complex world has no clear answer for a bounded, forgetful, under-sensing learner—the honest limit the hidden-structure premise of §2 implies.

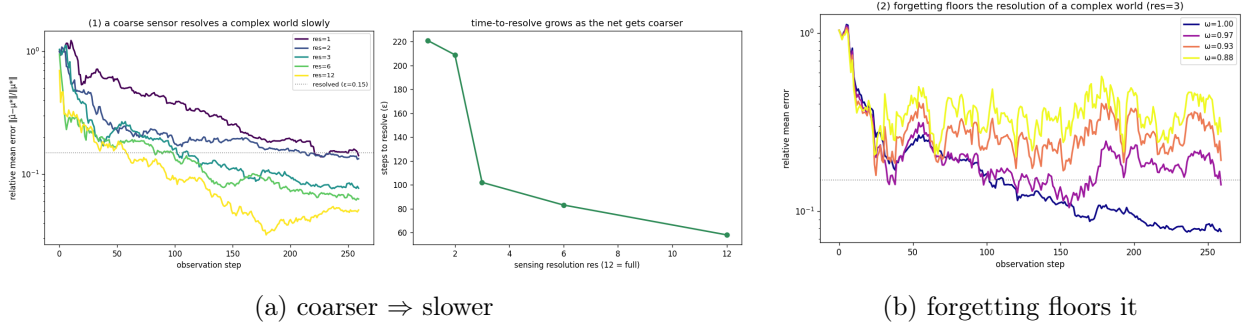


Figure 9: A coarse net on a complex world. a Relative error in the reconstructed world state over time, by sensing resolution; time-to-resolve grows sharply as the net coarsens. b With a fixed coarse sensor, forgetting plateaus the error at a floor that rises as ω falls—the world is never fully resolved.

F The staircase negative in detail

In the base dynamics the conservatism κ is a static read-out that never enters the update. The natural fix is to let κ gate the *rate* of revision: a per-node deposit gate $g_i = 1/(1 + \beta\kappa_i)$, PD-preserving, so a high- κ core node integrates evidence slower than a cheap belt node. We tested it on the phlogiston net, and it does *not* recover the staircase (Fig. 10). The belt $\rightarrow$ core lag stays negative (-55 steps at $\beta = 0$, -45 at large β): the high- κ nodes reorganize *first*, not last, and the κ -gradient gate is indistinguishable from a gradient-free uniform slowdown (-45) and a shuffled- κ gate (-45). Two reasons, which share one root cause. First, a deposit gate scales both numerator and denominator of the mean $\mu = \Pi^{-1}h$, so it barely throttles mean-tracking—and mean-tracking is

where theory identity lives. Second, the nodes that actually reorganize are exactly the high- κ mass-law commitments the relational operator refutes directly, while the low- κ belt are agreement nodes with nothing to reorganize, so the conservatism stratification is confounded with move-versus-don't-move. The root cause: with an operator-set Fisher deposit $J = H^\top H / \sigma^2$ —set by the observation operator, not the data—the structure does not move, so theory identity rides the posterior *means* and neither a rate gate on those means nor the stratified couplings can express the staircase. The negative localizes the missing ingredient to a data-dependent observation model in which the couplings themselves are learned (a structure-matched or nonlinear operator).

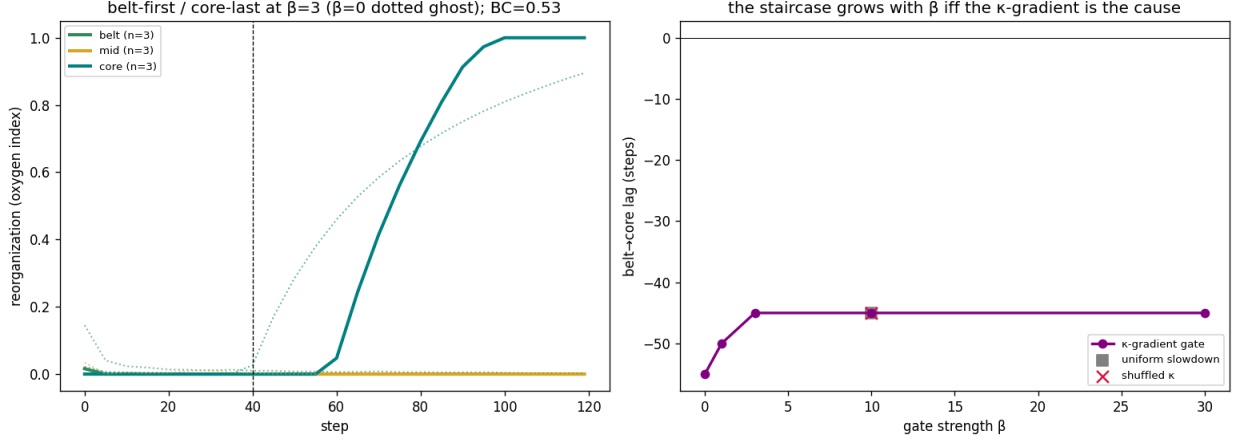


Figure 10: The staircase is *not* recovered by a conservatism-gated revision rate (phlogiston). *Left*: reorganization by κ -shell at the best β ($\beta = 0$ baseline dotted)—the high- κ “core” moves first and the low- κ “belt” (agreement nodes) never reorganizes, the reverse of the prediction. *Right*: the belt→core lag stays negative and the κ -gradient gate (purple) is matched by the gradient-free uniform (grey square) and shuffled- κ (red cross) nulls—the gradient adds nothing. The negative localizes the missing ingredient to data-dependent structure learning.